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An Interpretable Rule-Based Loan Approval System Using Hill Climbing Optimization

Abstract—Loan approval is a critical decision-making process in financial institutions, where accurate assessment of customer data is essential to minimize risk and ensure profitability. Traditional approaches predominantly rely on machine learning models such as decision trees, logistic regression, and ensemble techniques. Although these models achieve high predictive accuracy, they often lack interpretability, making it difficult for stakeholders to understand decision reasoning.

This paper proposes a heuristic search-based approach for generating optimized and interpretable decision rules for loan approval. The method employs the Hill Climbing algorithm to iteratively explore the solution space and improve candidate rules using a lift-based evaluation function that combines precision and coverage.

The system is evaluated on the Bank Marketing dataset from the UCI Machine Learning Repository. Experimental results demonstrate that the proposed approach generates meaningful decision rules and achieves improved performance compared to baseline methods, while maintaining transparency. The generated rules also provide valuable insights into customer behavior, particularly highlighting the importance of financial attributes such as account balance and age. The proposed method achieves a higher lift score compared to baseline approaches, demonstrating improved effectiveness in identifying potential loan subscribers.

Overall, the proposed method offers a flexible and interpretable framework for rule-based optimization, making it suitable for real-world financial decision systems.

Index Terms—Loan approval, Hill Climbing, heuristic search, interpretable AI, rule-based system

I. INTRODUCTION

Artificial Intelligence (AI) has become a fundamental component in modern decision-making systems, particularly in domains such as healthcare, finance, and e-commerce [7]. In the financial sector, loan approval is a critical application where institutions evaluate customer data to determine creditworthiness.

Traditional loan approval systems rely on machine learning models such as decision trees, logistic regression, and ensemble techniques [1], [2]. While these models achieve high predictive accuracy, they often lack interpretability, making it difficult for stakeholders to understand the reasoning behind decisions. This limitation is particularly significant in financial applications, where transparency and accountability are essential for regulatory compliance and user trust.

Another limitation of conventional approaches is their static nature. Once trained, these models do not dynamically adapt to new data patterns or explore alternative decision strategies. This restricts their effectiveness in real-world environments where data distributions may evolve over time.

To address these challenges, this paper proposes a rule-based loan approval system using heuristic search techniques.

Specifically, the Hill Climbing algorithm is employed to iteratively generate and optimize decision rules [7]. Unlike traditional machine learning models, the proposed approach focuses on exploring the solution space and refining rules based on an evaluation function, thereby enabling transparent and interpretable decision-making.

The proposed system enhances interpretability by producing human-readable rules and provides a flexible framework for dynamic optimization. This makes it more suitable for real-world financial decision systems where both performance and transparency are required.

II. RELATED WORK

Loan approval and credit risk prediction have been widely studied using machine learning techniques. Traditional models such as decision trees, logistic regression, and ensemble methods have demonstrated strong predictive performance in financial applications [1], [2]. These approaches analyze customer financial and demographic attributes to determine creditworthiness.

Recent research has focused on improving the interpretability of such models through explainable artificial intelligence (XAI) techniques. These methods provide post-hoc explanations for model predictions, helping to improve transparency in decision-making systems. However, they still rely on underlying black-box models and do not inherently generate interpretable decision rules.

Several studies have also explored the application of machine learning models for credit scoring and loan approval [3]. These systems achieve high accuracy by leveraging ensemble learning and advanced optimization techniques. Despite their effectiveness, they often lack transparency and fail to provide clear reasoning behind decisions, which is a critical requirement in financial systems.

To address these challenges, research in explainable AI emphasizes the need for transparent and accountable decision-making frameworks [4]. While these approaches improve interpretability, they remain dependent on static trained models and do not support dynamic exploration of alternative decision strategies.

In addition, recent studies in financial explainable AI highlight the growing importance of interpretability in real-world applications [5], [6]. Despite these advancements, most approaches focus on explaining model outputs rather than generating interpretable decision logic.

In contrast, classical artificial intelligence techniques such as heuristic search algorithms offer a different perspective. Hill

Climbing, a local search method, iteratively improves candidate solutions by exploring neighboring states and selecting better alternatives [7]. Unlike conventional machine learning approaches, heuristic search focuses on solution generation and refinement rather than learning fixed patterns from data.

A comparison with existing XAI-based approaches highlights a key limitation: most systems rely on post-hoc explanations rather than generating inherently interpretable decision rules. The proposed work addresses this limitation by adopting a rule-based framework combined with heuristic search. By using the Hill Climbing algorithm to dynamically generate and optimize decision rules, the system provides both interpretability and adaptability.

Overall, the literature indicates that while machine learning and explainable AI techniques have improved prediction performance and transparency, there remains a significant gap in developing systems that can generate and optimize interpretable decision rules. This motivates the proposed approach, which integrates heuristic search with rule-based modeling for loan approval.

III. RESEARCH GAP AND MOTIVATION

Despite significant advancements in machine learning for loan approval and credit risk assessment, several limitations remain in existing approaches. Traditional models such as Random Forest, Gradient Boosting, and Logistic Regression achieve high predictive accuracy, but they often operate as black-box systems, limiting their interpretability [1], [2]. This lack of transparency is a critical concern in financial applications, where decision-making must be explainable and compliant with regulatory requirements.

Recent research has attempted to address this issue through explainable artificial intelligence (XAI) techniques. These approaches provide post-hoc explanations for model predictions, improving transparency to some extent [4], [5]. However, they still rely on underlying machine learning models and do not inherently generate interpretable decision rules. As a result, the decision-making process remains dependent on complex models, and explanations are provided only after predictions are made.

Another key limitation of existing systems is their static nature. Most machine learning models are trained once and do not dynamically adapt to new data patterns or explore alternative decision strategies. This restricts their ability to optimize decision-making processes over time. Furthermore, current approaches rarely focus on rule-based reasoning, which is essential for producing human-readable and easily interpretable decisions. Issues such as class imbalance also remain inadequately addressed in many practical implementations [3].

A comparison with recent XAI-based loan approval systems highlights this gap. While these systems improve interpretability through model explanation techniques, they do not provide mechanisms for generating or optimizing decision rules dynamically [6]. In contrast, the proposed approach adopts a rule-based framework combined with heuristic search. By utilizing the Hill Climbing algorithm, the system explores the

solution space of possible rules and iteratively refines them based on an evaluation function [7].

This shift from model-based explanation to rule-based optimization addresses both interpretability and adaptability. The proposed method not only produces transparent and human-readable decision rules but also enables continuous improvement through dynamic optimization. Consequently, it provides a more effective and practical solution for real-world loan approval systems, where both accuracy and explainability are essential.

IV. SYSTEM ARCHITECTURE

As shown in Fig. 1, the proposed system follows a pipeline-based architecture designed to generate and optimize interpretable decision rules for loan approval. The architecture consists of sequential stages including data input, preprocessing, rule generation, optimization, evaluation, and output generation. Each stage performs a specific function and contributes to the overall objective of rule-based decision optimization.

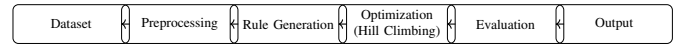


Fig. 1. Proposed system architecture for rule-based loan approval using Hill Climbing optimization.

Initially, the dataset is loaded from the Bank Marketing dataset, which contains customer demographic and financial attributes. The preprocessing stage transforms raw data into a suitable format by handling missing values, selecting relevant features such as age, balance, job, and marital status, and preparing the data for rule evaluation.

The rule generation module creates initial decision rules that serve as candidate solutions in the search space. Each rule is represented as a set of conditions applied to dataset attributes, for example: *IF balance > 2500 AND age > 55 THEN Approve Loan*.

The optimization stage applies the Hill Climbing algorithm to iteratively improve these rules. Neighboring rules are generated by modifying conditions such as adding or removing attributes or adjusting threshold values. Each candidate rule is evaluated using a scoring function, and the best-performing rule is selected for further refinement until convergence is achieved [7].

The evaluation module measures rule quality using metrics such as precision, coverage, and lift. These metrics ensure that the rules are both accurate and applicable to a significant portion of the dataset, while also addressing class imbalance.

Finally, the output module presents the optimized decision rules in a human-readable format. Unlike traditional machine learning models, the proposed system produces transparent and interpretable rules, enabling better understanding and decision-making in financial applications.

V. DATASET AND PREPROCESSING

The proposed system is evaluated using the Bank Marketing dataset obtained from the UCI Machine Learning Repository.

The dataset consists of 45,211 records with customer demographic and financial attributes, along with a binary target variable indicating loan subscription.

A subset of relevant features, including age, balance, job, and marital status, is selected to capture key aspects of customer behavior while maintaining simplicity and interpretability in rule generation.

The dataset exhibits a significant class imbalance, where non-subscribers greatly outnumber subscribers. This imbalance

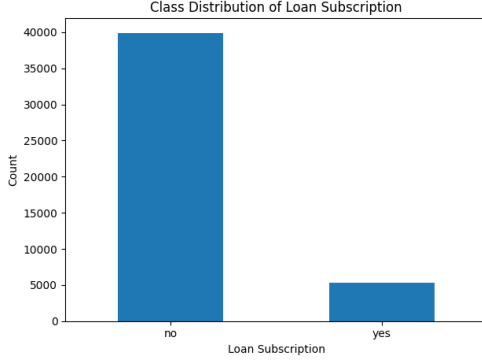


Fig. 2. Class distribution of loan subscription dataset

ance poses challenges for traditional models, which may become biased toward the majority class. To address this issue, a lift-based evaluation approach is incorporated along with precision and coverage metrics.

Data preprocessing is performed to ensure the quality and usability of the dataset. This includes handling missing values, removing duplicate records, and eliminating inconsistencies. Feature selection is applied to reduce complexity and focus on attributes that significantly influence loan approval decisions.

Categorical attributes such as job and marital status are directly used in rule expressions to preserve interpretability, while numerical attributes such as age and balance are transformed into threshold-based conditions. These transformations enable flexible rule representation and support efficient optimization.

Overall, preprocessing ensures that the dataset is clean, consistent, and suitable for rule-based learning, forming a strong foundation for generating accurate and interpretable decision rules.

VI. PROPOSED METHODOLOGY

The proposed system formulates loan approval as a rule optimization problem and employs a heuristic search approach to generate interpretable decision rules. Unlike traditional machine learning models that learn static patterns from data, this method dynamically explores the solution space and refines candidate rules using the Hill Climbing algorithm.

A. Rule Representation and Problem Formulation

Each decision rule is represented as a set of conditions applied to dataset attributes. A rule can be expressed in the form:

IF condition₁ AND condition₂ THEN Approve Loan

The rule optimization process is modeled as a state-space search problem, where each state corresponds to a candidate rule. The initial state is generated using selected features, and the objective is to identify a rule that maximizes an evaluation function.

B. Hill Climbing Optimization

Hill Climbing is employed as the core optimization technique to iteratively improve candidate rules [7]. Starting from an initial rule, neighboring rules are generated by modifying conditions, such as adding or removing attributes or adjusting threshold values. At each iteration, the algorithm selects the neighboring rule with the highest evaluation score and continues the process until no further improvement is achieved.

C. Evaluation Metrics

The quality of each rule is assessed using multiple evaluation metrics to balance accuracy and generality. Precision measures the correctness of predictions, while coverage represents the proportion of dataset instances that satisfy the rule. To address class imbalance, a lift-based measure is used to compare rule performance against baseline probability.

$$Precision = \frac{TP}{Total\ Predictions} \quad (1)$$

$$Coverage = \frac{Matched\ Samples}{Total\ Samples} \quad (2)$$

$$Lift = \frac{Precision}{Baseline} \quad (3)$$

The final score of a rule is computed as:

$$Score = 0.7 \times Lift + 0.3 \times Coverage \quad (4)$$

D. Optimization Process

The overall optimization process involves generating an initial rule, evaluating it using the scoring function, and iteratively refining it through neighbor exploration. The process continues until convergence, resulting in an optimized and interpretable decision rule.

The proposed approach enables efficient exploration of the search space while maintaining transparency, making it suitable for financial decision-making systems.

VII. EXPERIMENTAL RESULTS AND DISCUSSION

This section presents the performance of the proposed rule-based loan approval system optimized using the Hill Climbing algorithm. The system generates interpretable decision rules and evaluates them using a lift-based scoring mechanism.

A. Generated Decision Rules

The proposed approach produces meaningful decision rules that capture patterns in customer behavior. Examples of optimized rules include:

- **Rule 1:** IF balance > 2500 AND age > 55 THEN Approve Loan
- **Rule 2:** IF job = retired THEN Approve Loan

These rules demonstrate the ability of the system to identify significant factors influencing loan subscription decisions.

B. Performance Evaluation

The performance of the generated rules is assessed using precision, coverage, and lift. The lift metric evaluates the effectiveness of rules compared to baseline probability, ensuring robustness in the presence of class imbalance.

The results indicate that the optimized rules achieve higher performance than random selection, confirming the effectiveness of the heuristic search approach. The combination of lift and coverage ensures that the rules are both accurate and applicable to a meaningful portion of the dataset.

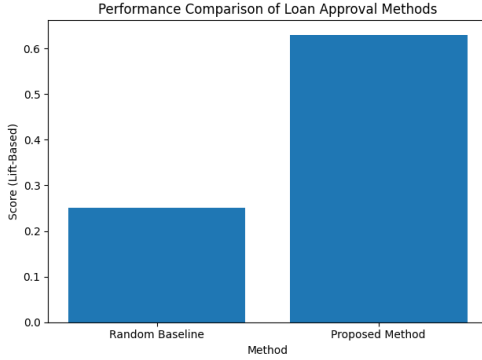


Fig. 3. Performance comparison between baseline and proposed method

C. Convergence Analysis

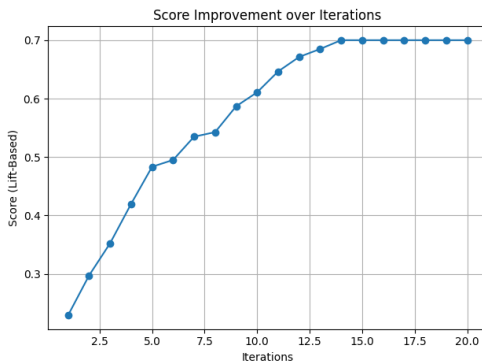


Fig. 4. Score improvement over iterations

The convergence behavior of the Hill Climbing algorithm is illustrated in Fig. 4. The evaluation score increases rapidly during initial iterations, indicating efficient exploration of the

solution space. As the algorithm progresses, improvements become gradual and eventually stabilize, reflecting convergence to a locally optimal solution.

D. Discussion

The results demonstrate that the proposed system effectively generates interpretable and optimized decision rules. Unlike traditional machine learning models that operate as black-box systems, the rule-based approach provides transparent outputs that can be directly understood by stakeholders.

However, several challenges influence the performance of the system. One major challenge is the large search space associated with rule generation, as multiple attributes and threshold combinations lead to a vast number of possible rules. Although the Hill Climbing algorithm reduces computational complexity through local search, efficient neighbor generation remains essential.

Another limitation is the tendency of Hill Climbing to converge to local optima. Since the algorithm selects the best neighboring solution at each step, it may fail to explore globally optimal rules. Techniques such as random restarts or introducing stochastic variations can help improve exploration.

The dataset also presents class imbalance, where non-subscribers significantly outnumber subscribers. To address this, a lift-based evaluation metric is employed, ensuring that the generated rules remain effective in identifying minority class instances.

Additionally, there exists a trade-off between interpretability and performance. While simpler rules improve readability, more complex rules may yield better predictive performance. Balancing these aspects is crucial for practical deployment.

Despite these challenges, the proposed approach demonstrates strong potential by combining interpretability with dynamic optimization, making it suitable for real-world financial decision-making systems.

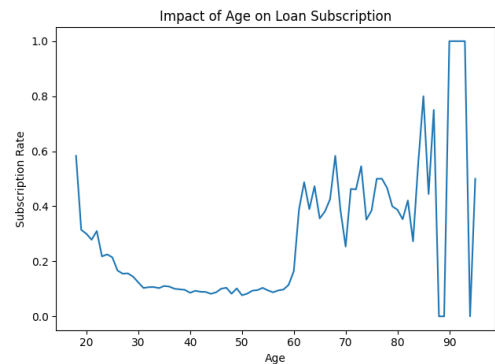


Fig. 5. Impact of age on loan subscription

VIII. CONCLUSION

This paper presented a rule-based approach for loan approval prediction using the Hill Climbing algorithm. The proposed system formulates decision-making as an optimization problem and focuses on generating interpretable and optimized

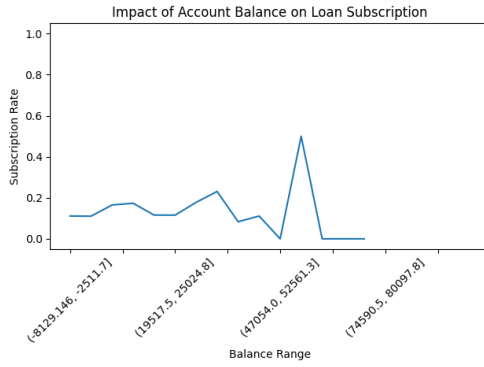


Fig. 6. Impact of account balance on loan subscription

decision rules, addressing the limitations of traditional machine learning models that operate as black-box systems. The experimental results validate the effectiveness of the proposed approach on a real-world dataset consisting of 45,211 records.

The approach was evaluated using the Bank Marketing dataset, where relevant features such as age, balance, job, and marital status were selected to support effective rule generation. A lift-based evaluation mechanism was employed to handle class imbalance and ensure that the generated rules are both accurate and applicable to a meaningful portion of the dataset.

The experimental results demonstrate that the proposed method successfully identifies significant patterns in customer behavior and generates transparent decision rules. The convergence characteristics of the Hill Climbing algorithm show efficient exploration of the solution space, leading to improved rule quality over successive iterations.

In contrast to existing explainable AI approaches that rely on post-hoc explanations such as SHAP and LIME, the proposed system provides inherent interpretability by directly generating human-readable rules. Additionally, the use of heuristic search enables dynamic rule optimization, allowing the system to adapt and refine decision logic without relying on static trained models.

Overall, the proposed framework offers a practical and flexible solution for financial decision-making systems, where both interpretability and performance are critical. The study demonstrates that heuristic search-based rule optimization can serve as an effective alternative to conventional machine learning techniques, particularly in applications requiring transparency and accountability.

IX. FUTURE WORK

Future work can explore the integration of heuristic search with machine learning models to improve predictive performance while maintaining interpretability. Advanced optimization techniques such as Genetic Algorithms or Simulated Annealing can be investigated to overcome the limitations of Hill Climbing and improve exploration of the solution space.

The inclusion of additional features and more complex datasets can further enhance rule quality and model robustness.

Techniques for handling class imbalance, such as resampling or cost-sensitive learning, can also be incorporated.

Furthermore, the proposed approach can be extended to other domains such as credit risk assessment, fraud detection, and insurance analytics, demonstrating its applicability to a wide range of decision-making problems.

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